RMIT Classification: Trusted

ISYS3416/3432 – Software Engineering Fundamentals for IT
Tutorial 3

Part A: User Story.

Identify the category of the following user stories and determine if the user story is ready for the Sprint.

1. As an online shopper, I want to add items to my wishlist so that I easily find and buy them in the future.
2. As a developer, I want to implement blockchain-based authentication so that I can enhance system security and ensure user credentials are tamper-proof.
3. As a user, I want a mobile app version of the website with synchronized logins so I can easily switch between devices without signing in repeatedly.
4. As a user, I want to reset my password via email verification so that I can securely regain access to my account.
5. As a user, I want to view my account balance on the dashboard so that I can easily check my balance.

Part B: Use Case

1. From the given scenario, create the use case diagram and the appropriate relationships (include, extend)
   Janet: When we are selling insurance, the first thing to do is to get the details of the member: their age and occupation, where they live, and their insurance history (any recent accident). If customers are not sure of the policy they want, we recommend the ones that best meet their needs.

   Perez: Where does the customer information come from?

   Janet: It may be from the system or collected over the phone.

   Perez: How do you choose the right policy?

   Janet: We'll try to find the best policy search by searching for the most appropriate one based on the information provided.

2. Draw the use case diagram of the given scenario

   The Online Library System allows students to search for books, borrow books, and return them. The librarian manages book records, approves borrowing requests, and updates book availability. The administrator maintains user accounts.

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Part C: Functional or Non-Functional Requirements.

Identify if the statement is functional requirements (FR) or non-functional requirements (NFR)

1. The Mentcare system shall run on hardware (Linux servers) that is available in the authority's data centre. Note that the maximum server memory available is currently 32GB. System administrators in the center are responsible for system backup.
2. The Mentcare system shall include a search feature that allows users to discover the records for individual patients. Search may be based on patient name or the patient's national health identifier.
3. The Mentcare system shall be continuously available during 'normal' clinic working hours between 0800 and 1830, Monday to Friday.
4. It shall be possible to update a patient record during a consultation when the record has been opened or at a later date. Records that have not been updated during a consultation should be flagged to indicate that they are not completely up-to-date.
5. The Mentcare system shall make use of the Mid-Scotland health authority's single sign-on authentication system.
6. Periods of scheduled maintenance for the Mentcare system shall normally be arranged during the authority's "systems at risk" periods i.e. between 2100 and midnight, Monday to Friday, and between 0800 and 1200, Sunday.

Part B: Activity

Imagine you (students) are a software engineer working in a software development organization and want to develop an Online Bookstore Platform. The Online Bookstore Platform stores and sells different types of books. In the following parts, you are expected to work on the requirements engineering process of the Online Bookstore Platform. You have freedom and have to add more features, services, requirements, etc. to the Online Bookstore Platform.

A. Stakeholders of the Online Bookstore Platform

1. Identify **TWO** stakeholders for the Online Bookstore Platform using the following template.
2. Discuss which of those stakeholders are end-users of the Online Bookstore Platform.

Stakeholder Name
Description: A short description of the goal, services used/provided by, or responsibility of the stakeholder.

B. User and System Requirements of the Online Bookstore Platform

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1. Write **ONE** user requirement for the Online Bookstore Platform and then write **TWO** system requirements based on the user requirement.

C: Functional Requirements of the Online Bookstore Platform (15 mins)

1. Identify ONE functional requirement for each stakeholder of the Online Bookstore Platform using the following template.

Template The shall/should/must <function>

D. Non-functional Requirements of the Online Bookstore Platform (20 mins)

1. Identify and document TWO to FOUR non-functional requirements of the Online Bookstore Platform using the following template.
2. Discuss which of these non-functional requirements are the most important for the Online Bookstore Platform.
3. Discuss which of these non-functional requirements are (more) important for the identified stakeholders.

Template The shall/should/must <function>